Experiment No. 3

Analysis of UV-VIS spectra of various compounds

OBJECTIVE:

Most of the conjugated compounds are not transparent in the portions of the electromagnetic spectrum which we call as ultraviolet (UV) and visible (Vis) regions, that is, the region where wavelengths range from 190 nm to 800 nm. The objective is to (i) verify the validity of Beer-Lambert law using various aqueous solutions of benzoic acid and determination of unknown concentration of benzoic acid; (ii) study the effect of the substituent group present in benzene; (iii) study the effect of the acid and base on the substituted benzene

ORIGIN:

For an atom that absorbs in the ultraviolet, the absorption spectrum sometimes consists of very sharp lines, as would be expected for a quantized process occurring between two discrete energy levels. For molecules, however, the UV absorption usually occurs over a wide range of wavelengths, because molecules (as opposed to atoms) normally have many excited modes of vibrations and rotation at room temperature. In fact, the vibration of molecules cannot be completely "frozen out" even at absolute zero. Consequently, a collection of molecule generally has its members in many states of vibrational and rotational excitation. The energy levels for these states are quite closely spaced, corresponding to energy differences considerably smaller than those of electronic levels. The rotational and vibrational levels are thus "superimposed" on the electronic levels. A molecule may therefore undergo electronic and vibrational-rotational excitation simultaneously, as shown in Figure 1.

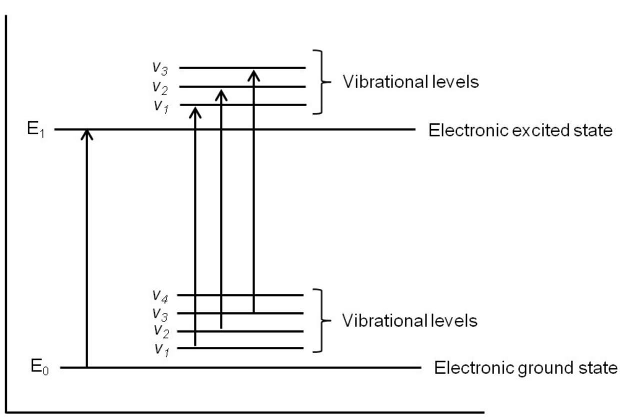

Figure 1. Electronic transitions with vibrational transitions superimposed.

Because there are so many possible transitions, each differing from the others by only a slight amount, each electronic transition consists of a vast number of lines spaced so closely that the spectrophotometer cannot resolve them. Rather, the instrument traces an "envelop" over the entire pattern. What is observed from these types of combined transitions is that the UV spectrum of a molecule usually consists of a broad band of absorption centered near the wavelength of the major transition.

PRINCIPLE:

The greater the number of molecules capable of absorbing light of a given wavelength, the greater the extent of light absorption is possible. Furthermore, the more effectively a molecule absorbs light of a given wavelength, the greater the extent of light absorption. From these guiding ideas, the following empirical expression, known as the **Beer-Lambert's law**, may be formulated.

$$A = \log(I_0/I) = \varepsilon c l \text{ for a given wavelength}$$

A = absorbance, I_0 = Intensity of light incident upon sample cell

I = Intensity of light having sample cell C = molar concentration of solute

l = length of sample cell (cm) ε = molar absorptivity

The term $\log(I_0/I)$ is also known as the **absorbance** (or the **optical density** in older literature) and may be represented by A. The molar absorptivity (formerly known as molar extinction coefficient) is a property of the molecule undergoing an electronic transition and is not a function of the variable parameters involved in preparing as solution. The size of the absorbing system and the probability that the electronic transition will take place control the absorptivity, which ranges from 0 to 10^6. Values above 10^4 are termed high-intensity absorptions, while values below 10^3 are low intensity-absorptions. Forbidden transitions have absorptivities in the range from 0 to 1000.

The Beer-Lambert law is rigorously obeyed when a single species give rise to the observed absorption. The law may not be obeyed, however, when different forms of the absorbing molecule are in equilibrium, when solute and solvent form complexes through some sort of association, when thermal equilibrium exists between the ground electronic state and a low line excited state, or when fluorescent compounds or compounds changed by irradiation are present.

Beer and Lambert's law states that the absorbance of a solution containing light absorbing material depends on the follow factors:

1. The nature of the substance 2. The wavelength of light

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3. The path of light
4. The amount of colored material in the light path

This experiment also aims to show the effect of different substituent groups attached with benzene ring on their electronic transitions as nature (electron donating, electron withdrawing etc.,) of the substituent attached with the ring influences the absorption spectrum significantly. This experiment shows the effect in benzene, by comparing the absorption spectra of benzene, benzoic acid and 4-aminobenzoic acid.

REAGENTS: Benzene, Benzoic acid**,** 4-aminobenzoic acid, Hydrochloric acid (0.1 M), Sodium hydroxide solution (0.1 M), Methanol

APPARATUS: 250 ml volumetric flask (2), 100 mL volumetric flasks (7), 10 mL volumetric flask (5), 50 mL volumetric flask (1), graduated pipette 10 mL (2), wash bottle, 250 mL beaker (2) cylinder

EXPERIMENTAL PROCEDURE:

To verify Beer-Lambert's law, a set benzoic acid solution of different concentrations prepared and spectra are recorded. In this experiment, the effect of the substituent group present in benzene is observed by comparing the absorption spectra of benzene, benzoic acid and 4-aminobenzoic acid and also dissolved in dilute acid and basic solution.

(**1)** Take 0.2 mL benzene in 50 mL volumetric flask and prepare a solution by diluting to the mark with methanol (solution A_1). Using Stoppard quartz cells, record the absorption spectrum of the solution over the wavelength range 210-400 nm using pure methanol as the blank.

(2) Prepare a solution of benzoic acid in distilled water by dissolving 0.1 gm in 250 mL volumetric flask and making up to the mark (A_2). Now take 1 mL of solution A_2 and dilute into 10 mL volumetric flask with distilled water (B_2). Take 2 mL, 4 mL, 5 mL, 8 mL and 12 mL of solution A_2 in 5 different 100 ml volumetric flasks and make the volume up to the mark. Record the spectra for all the 6 solutions and note the absorbance at λ_{max}. Draw the calibration curve and show the validity of Beer Lambert law and also calculate the slope of the graph.

(3) Prepare similar solution of 4-aminobenzoic acid (A_3) in distilled water by taking 0.1 gm 4-aminobenzoic acid in 250 mL. Prepare the similar solution of B_3 from solution A_3 (take 1 mL of solution A_3) and record the absorption spectrum of the solution (B_3) over the wavelength range of 210 - 400 nm, using distilled water as blank

(4) Now prepare a new solution of 4-amino benzoic acid (C_3) by placing 1 mL of solution A_3 in a 10 ml flask and make up the volume with 0.1 (M) NaOH solution. Record the absorption spectrum.

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(5) Now prepare a solution of 4-amino-benzoic acid (D_3) by placing 1 mL of solution A_3 in a 10 mL flask and make up the volume with 0.1 (M) HCl solution. Record the absorption spectra for D_3.

OBSERVATIONS:

Various solution	Conc.	Name of the solution	Wavelength (nm)	Absorbance
A_1		Benzene		
B_2	1 Ml	Benzoic acid (BA)		
B_2	2 Ml			
B_2	4 Ml			
B_2	5 Ml			
B_2	8 Ml			
B_2	12 Ml			
B_3	1 Ml	4-aminobenzoic acid (ABA)		
C_3	1 Ml	4-ABA in NaOH		
D_3	1 Ml	4-ABA in HCl		

Examine the six absorption spectra A_1, B_2, B_3, C_2, C_3, D_2, and D_3 and record the λ_{max} values of the absorption peaks. Comment on how –COOH and $–NH_2$ groups affect the absorption spectrum of benzene and how HCl and NaOH affect the spectra.

CAUTION:. Never touch or smell directly any of the chemicals. Handle them carefully with spatula, burette, measuring cylinders etc. as per advice. By any chance if you come into contact with any of these chemicals, immediately wash with plenty of water and report.

NOTES: